PROJECT DEMONSTRATION & DOCUMENTATION

Food Ordering Behaviour and Consumer Trends:

A Structured Analysis of Choices and Habits

Prepared using Tableau

Data Analytics Project

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1. Introduction / Project Overview

The food delivery industry has grown into a highly competitive space where businesses must continually understand customer behaviour, streamline delivery performance, and identify opportunities to maximise revenue. While companies in this space generate large volumes of transactional data, translating that raw data into clear, actionable insight remains a persistent challenge.

This project addresses that challenge by analysing food ordering data using Tableau to uncover meaningful patterns and trends. The dataset used in the analysis includes key attributes such as Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, and Time Taken for Delivery. By building interactive Tableau dashboards on top of this data, the project enables a deeper understanding of how different factors influence order volume, customer preferences, and operational performance.

Through this analysis, the project surfaces critical insights such as top-performing cities, high-demand cuisines, peak mealtimes, and delivery time distribution — helping identify both areas of strong performance and opportunities for improvement.

2. Problem Statement

Despite having access to large volumes of order-level data, food delivery businesses often struggle to identify the key factors that influence order demand, customer spending patterns, and service efficiency. Without a structured analytical approach, valuable signals in the data — such as which cities are underperforming, which cuisines are trending, or where delivery delays are concentrated — remain hidden. This creates a clear need for a data-driven, visual approach that can convert raw operational data into insights that support faster and more confident business decisions.

3. Objectives of the Project

- To analyse food order data and identify patterns in customer ordering behaviour.
- To evaluate delivery performance and pinpoint operational bottlenecks.
- To identify high-demand cuisines, popular meal types, and peak ordering times.
- To compare order volumes and revenue contribution across different cities.
- To build an interactive Tableau dashboard that supports data-driven decision-making for Business Analysts, Operations Managers, and other stakeholders.
- To document the project comprehensively for demonstration, review, and future reproducibility.

4. Dataset Description

The dataset used for this analysis captures detailed information about individual food delivery orders. The key fields used across the dashboards are summarised below:

Field	Description
Order ID	Unique identifier assigned to each food order
Order Value	Total monetary value of the order placed by the customer
Delivery Fee	Fee charged for delivering the order to the customer
Cuisine	Type of food/cuisine ordered (e.g., Indian, Chinese, Italian, etc.)
Meal Type	Category of meal such as Breakfast, Lunch, Dinner, or Snacks
City	City in which the order was placed and delivered
Time Taken for Delivery	Duration between order placement and delivery completion

5. Tools & Technologies Used

Tool / Technology	Purpose
Tableau	Building interactive dashboards and visual analysis
Structured Dataset (CSV/Excel source)	Source data containing order, delivery, and customer attributes
Tableau Workbook / Public Dashboard	Hosting and sharing the final visual analysis
Documentation (this report)	Recording project scope, methodology, and outcomes for stakeholders

6. User Personas / Usage Scenarios

To ground the analysis in real stakeholder needs, the project is framed around three representative personas, each using the dashboard for a different purpose.

6.1 Rahul Sharma — Business Analyst

Rahul analyses order trends, cuisine preferences, and city-wise demand using Tableau dashboards. He studies metrics such as Order Value and Meal Type to identify high-performing segments and customer behaviour patterns. His analysis supports strategic, data-driven business decisions. He also tracks changes in demand across different time periods to understand evolving trends, enabling the business to align its offerings with customer needs and market dynamics.

6.2 Priya Mehta — Operations Manager

Priya focuses on improving delivery performance and overall operational efficiency. She analyses delivery time distribution and order volumes across different cities to identify delays and inefficiencies, with the goal of optimising resource allocation and ensuring faster deliveries. By monitoring peak hours and workload distribution, she helps streamline operations, reduce service gaps, and maintain high customer satisfaction.

6.3 Neha Reddy — Customer

Neha, a regular customer, values quick delivery, reasonable pricing, and a variety of cuisine options. She prefers ordering during peak meal times such as lunch and dinner, depending on convenience. Her satisfaction is largely influenced by timely delivery and overall service experience — when her expectations are met, she is more likely to reorder and remain loyal to the platform. Her behaviour reflects key customer trends that highlight the importance of quality, speed, and convenience in driving business success.

7. Methodology / Approach

1. Data Collection: Order-level data was gathered, covering order value, delivery fee, cuisine, meal type, city, and delivery time.
2. Data Preparation: The dataset was reviewed and cleaned to ensure consistency in formats (dates, currency values, city names, and categorical fields) before being connected to Tableau.
3. Exploratory Analysis: Initial exploration was carried out to understand distributions of order value, delivery time, and category frequencies (cuisine, meal type, city).
4. Dashboard Design: Interactive Tableau dashboards were designed around the three personas — Business Analyst, Operations Manager, and Customer-behaviour view — so each stakeholder can quickly access relevant insights.
5. Visualization Development: Charts including bar charts, line trends, heat maps, and geographic maps were built to represent order volume, cuisine demand, delivery performance, and city-wise comparisons.
6. Insight Extraction: Key findings were derived by interpreting the visualizations, focusing on top-performing cities, high-demand cuisines, peak mealtimes, and delivery time distribution.
7. Validation & Review: The dashboards and insights were reviewed for accuracy and relevance to the original business problem before being finalised for demonstration.

8. Dashboard Features & Functionality (Demonstration)

The Tableau dashboard is organised to support exploration from multiple angles. Its key features include:

- City-wise Order Analysis: A geographic/bar visualization comparing order volume and revenue contribution across cities, helping identify top-performing and underperforming markets.
- Cuisine Preference Analysis: Visualization ranking cuisines by order frequency and revenue, highlighting customer preferences and trending cuisine categories.

- Meal Type & Peak Time Analysis: Breakdown of orders by meal type (Breakfast, Lunch, Dinner, Snacks) and by time of day, revealing peak ordering windows.
- Delivery Time Distribution: Visualization showing how delivery times are distributed, used to flag delays and cities/segments with slower service.
- Order Value & Delivery Fee Trends: Analysis of average order value and delivery fee across cities, cuisines, and meal types to understand spending patterns.
- Interactive Filters: Filters for city, cuisine, meal type, and date range allow each persona to drill down into the specific view relevant to their role.
- KPI Summary Cards: At-a-glance metrics such as total orders, average order value, average delivery time, and top cuisine/city, giving stakeholders a quick operational snapshot.

Note: Insert dashboard screenshots here during final demonstration walkthrough (e.g., City-wise Orders, Cuisine Demand, Delivery Time Distribution, Peak Meal Times).

9. Key Insights & Findings

- Top-performing cities: Certain cities consistently generate a disproportionately higher share of total orders and revenue, indicating strong market demand that can guide expansion or marketing focus.
- High-demand cuisines: A small set of cuisine categories account for the majority of orders, suggesting where menu and partner-restaurant investment would have the greatest impact.
- Peak mealtimes: Lunch and dinner windows show the highest concentration of orders, aligning with expected customer behaviour and supporting more precise staffing and delivery-fleet planning.
- Delivery time distribution: While most orders are delivered within an acceptable time range, a visible subset of orders take noticeably longer — pointing to specific cities or time windows where operational improvement is needed.
- Order value patterns: Average order value varies across cities and cuisines, offering opportunities for targeted promotions or bundling strategies in lower-spend segments.
- Customer experience linkage: Faster, more consistent delivery correlates with the behaviour patterns valued by customers like Neha — reinforcing that delivery speed and consistency are directly tied to satisfaction and repeat ordering.

10. Outcomes & Business Value

The analysis translates raw order-level data into insights that are directly usable by different stakeholders. Business Analysts can use the findings to prioritise high-demand cuisines and markets; Operations Managers can use the delivery-time and workload analysis to reallocate resources and reduce delays; and the customer-behaviour perspective helps the business understand what drives loyalty and repeat orders. Collectively, this supports more informed, evidence-based decision-making across strategy and operations.

11. Reproducibility & Usability Notes

To ensure the project can be reviewed, reproduced, or extended by others, the following should be maintained alongside this document:

- The underlying dataset (or a clearly documented sample/schema) used to build the dashboards.
- The Tableau workbook file (.twbx) or a published Tableau Public/Server link.
- A GitHub repository link containing any data-preparation scripts, the dataset schema, and this documentation.
- A recorded or live demo walkthrough covering each dashboard view and persona use-case.

As noted in the project workspace, the team lead should update the Demo and GitHub links so that the mentor can access and evaluate the live dashboard and source files alongside this written documentation.

12. Conclusion

This project demonstrates how a structured, Tableau-driven analysis can turn raw food-delivery data into clear, actionable insight. By examining order value, delivery fee, cuisine, meal type, city, and delivery time through the lens of three real-world personas, the analysis highlights the factors that most strongly influence order demand, customer satisfaction, and operational efficiency. The resulting dashboards and insights provide a practical foundation for data-driven decisions that can help a food delivery business grow its top-performing segments while addressing operational gaps.

13. Future Scope

- Incorporating customer feedback/ratings data to correlate delivery performance directly with satisfaction scores.
- Adding predictive analysis (e.g., forecasting demand by city or cuisine) to support proactive planning.
- Expanding the dataset to include order cancellations and refunds for a fuller view of service quality.
- Automating dashboard refresh through a live data connection for real-time monitoring.

14. Appendix

14.1 Glossary

- Order Value — Total amount billed to the customer for an order, excluding/including delivery fee as defined in the dataset.
- Meal Type — Categorical grouping of orders by time-of-day meal occasion (Breakfast, Lunch, Dinner, Snacks).
- Time Taken for Delivery — Elapsed time from order placement to delivery completion, used as the primary operational-efficiency metric.